SPRINT 1

STAGGING TABLES:

- CDR_RAW_TELECOM
- CUSTOMER_MASTER
- TOWER_MASTER

CDR_RAW_TELECOM→MAP_STG_CDR

```
CREATE TABLE STG_CDR (
   Call_ID       INT,
   Caller        VARCHAR(15),
   Receiver      VARCHAR(15),
   Duration      INT,
   CallType      VARCHAR(10),
   TowerID       VARCHAR(10),
   Timestamp_Str  VARCHAR(50)
);
```

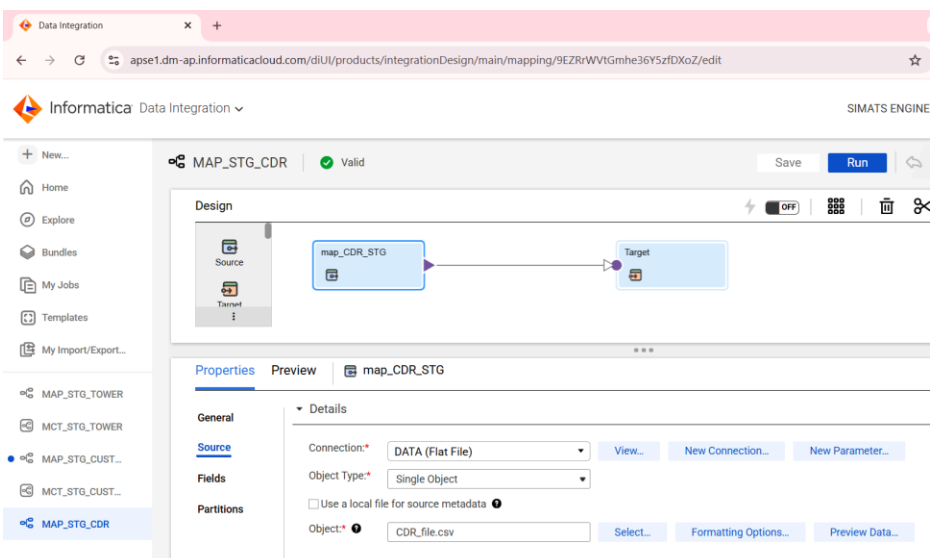


FIG1: "This mapping loads Call Detail Record data from a flat file *CDR_file.csv* into a staging table. It is a simple source-to-target mapping without transformations, used for initial data ingestion."

TOWER MASTER→STG_TOWER

```
CREATE TABLE STG_TOWER (
   TowerID   VARCHAR(10),
   Region    VARCHAR(50),
   City      VARCHAR(50)
);
```

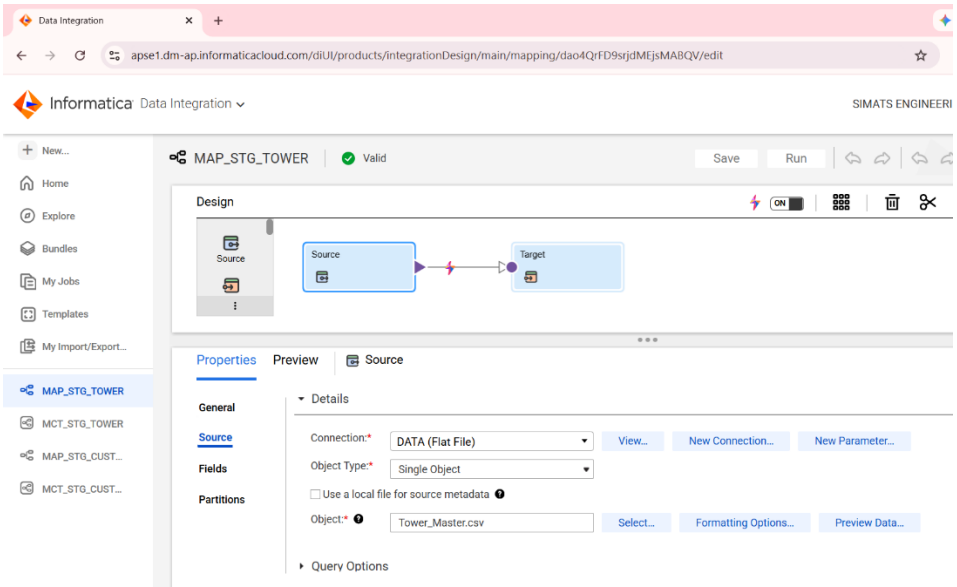


FIG2: "This mapping loads data from a flat file (Tower_Master.csv) into a staging table using a simple Source-to-Target flow without any transformations."

CUSTOMER_MASTER→MAP_STG_CUSTOMER

```
CREATE TABLE STG_CUSTOMER_HACKK (
   PhoneNumber    NUMBER(15),
   CustomerName   VARCHAR(100),
   PlanType       VARCHAR(20),
   TowerID        VARCHAR(10)
);
```

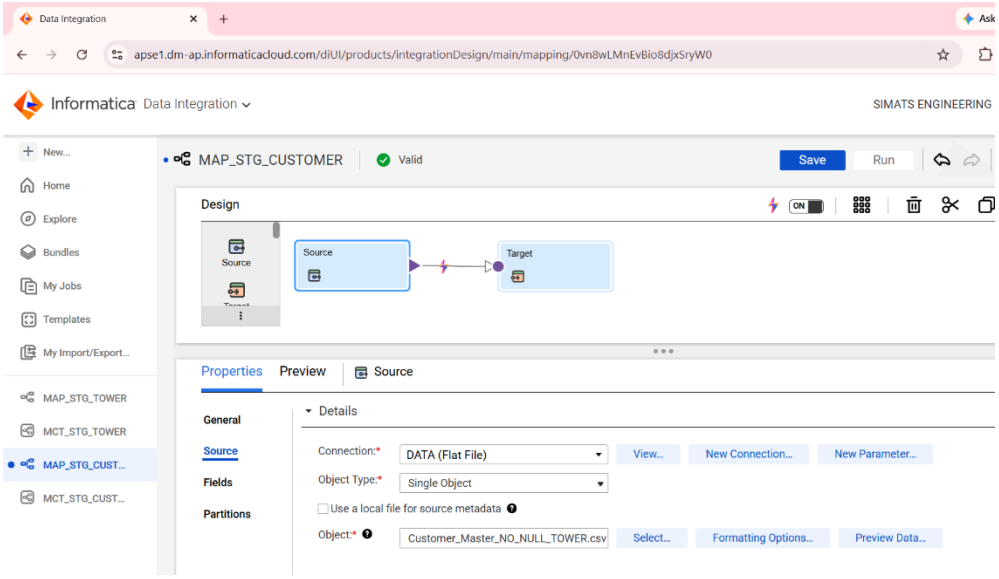


FIG 3: "This mapping loads customer data from a flat file *Customer_Master_NO_NULL_TOWER.csv* into a staging table. It is a simple source-to-target mapping without transformations, used for initial data loading."

DIMENSION TABLE

DIM_DATE

```
  CREATE TABLE DIM_DATE (
   DATE_KEY       NUMBER(8) PRIMARY KEY,
   FULL_DATE      DATE,
   DAY_NUM        NUMBER(2),
   MONTH_NUM      NUMBER(2),
   MONTH_NAME     VARCHAR2(20),
   QUARTER_NUM    NUMBER(1),
   YEAR_NUM       NUMBER(4),
   WEEK_DAY       VARCHAR2(20)
);
DECLARE
   v_start_date DATE := DATE '2024-01-01';
BEGIN
   FOR i IN 0..2190 LOOP   -- ~6 years
      INSERT INTO DIM_DATE (
         DATE_KEY,
         FULL_DATE,
         DAY_NUM,
         MONTH_NUM,
         MONTH_NAME,
         QUARTER_NUM,
         YEAR_NUM,
         WEEK_DAY
```

```
    )
    VALUES (
      TO_NUMBER(TO_CHAR(v_start_date + i, 'YYYYMMDD')),
      v_start_date + i,
      EXTRACT(DAY FROM v_start_date + i),
      EXTRACT(MONTH FROM v_start_date + i),
      TRIM(TO_CHAR(v_start_date + i, 'MONTH')),
      TO_NUMBER(TO_CHAR(v_start_date + i, 'Q')),
      EXTRACT(YEAR FROM v_start_date + i),
      TRIM(TO_CHAR(v_start_date + i, 'DAY'))
    );
  END LOOP;
  COMMIT;
END;
/
```

--Step 1: Create the Table

-- Step 2: Declare Start Date

```
DECLARE
  v_start_date DATE := DATE '2024-01-01';
```

-- Step 3: Begin PL/SQL Block

```
BEGIN
```

--Step 4: Loop Through Dates

```
  FOR i IN 0..2190 LOOP   -- ~6 years
```

Loop runs **2191 times (~6 years)**

- i = 0 → 01-Jan-2024
- i = 1 → 02-Jan-2024

- … continues day by day

--Step 5: Generate Each Date

- v_start_date + i,

--Step 6: Insert Data into Table

INSERT INTO DIM_DATE (...)

VALUES (...)

Step 7: Generate DATE_KEY

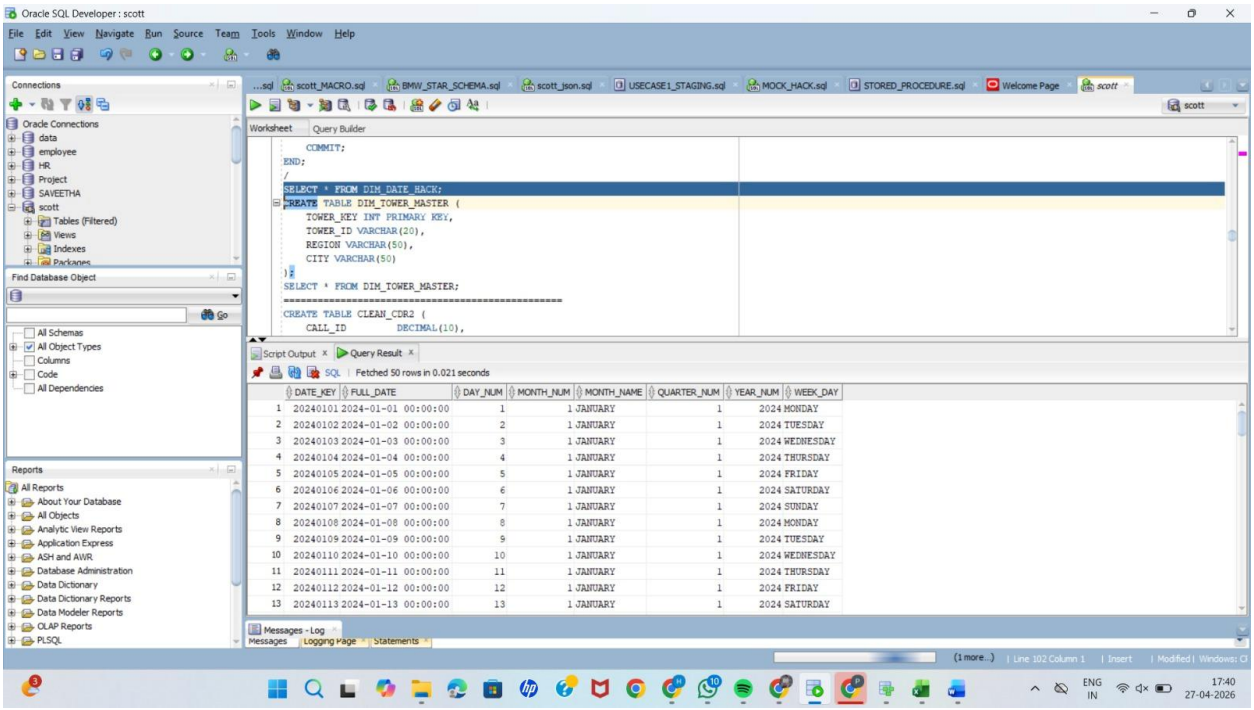
TO_NUMBER(TO_CHAR(v_start_date + i, 'YYYYMMDD'))

Step 8: Extract Day,month,year

EXTRACT(DAY FROM v_start_date + i)

EXTRACT(MONTH FROM v_start_date + i)

EXTRACT(YEAR FROM v_start_date + i)

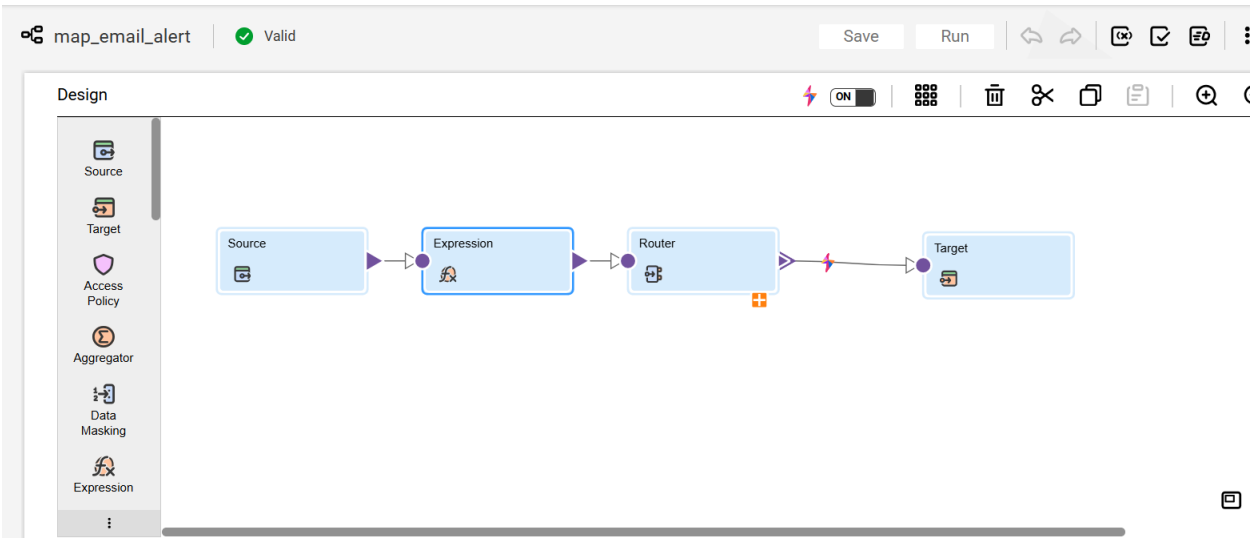


USE CASE--5

EMAIL ALERT NOTIFICATIONS

"In addition to core ETL, I implemented enhanced features like email alert notifications using an Email Task in the workflow, triggered conditionally after identifying high usage customers through expression and Router transformations."

MAPPING:



Expression:

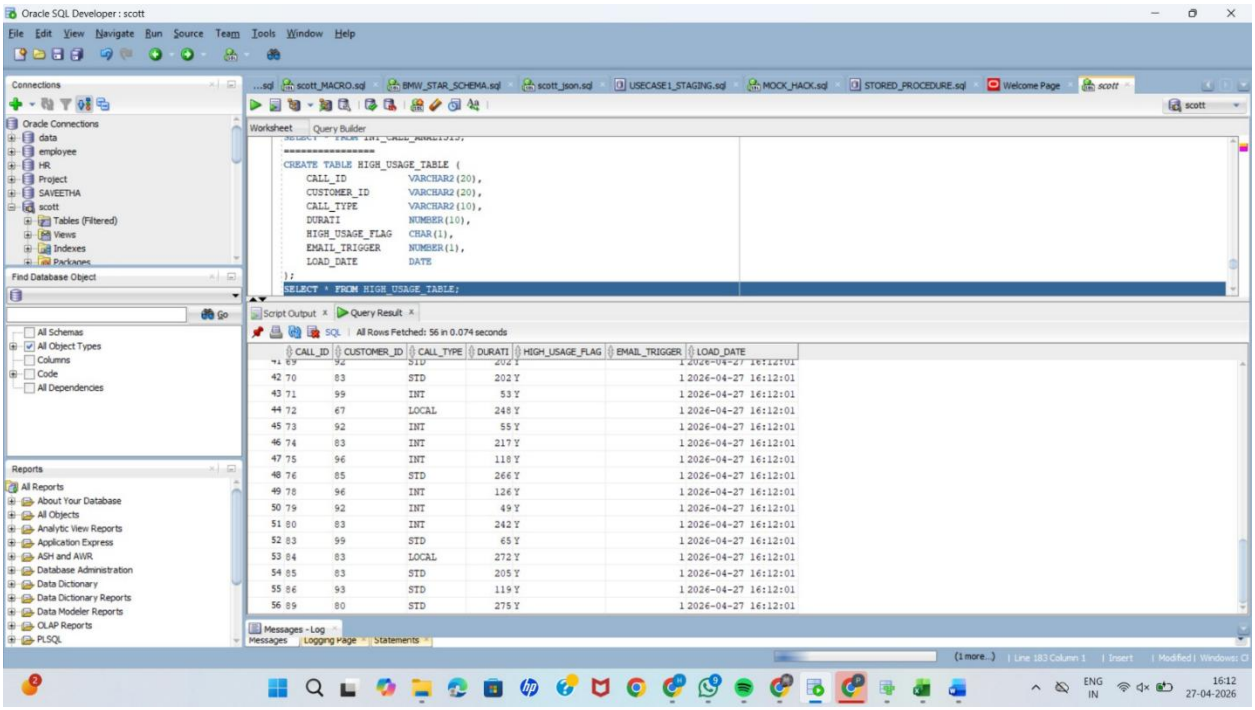
Create simple expressions. You can also use expression macros to create complex expressior

☐ Allow additional fields and expressions during task creation

Expressions

Field Name	Expression	Default Value	Type
HIGH_USAGE_FLAG	IIF(DURATI > 10, 'Y', 'N')	ERROR('transformation err...	string
EMAIL_TRIGGER	IIF(DURATI > 10, 1, 0)	ERROR('transformation err...	string
o_load_date	SYSDATE	ERROR('transformation err...	date/time

Router:

Output Groups

Group Name	Condition
DEFAULT1	
HIGHUSAGE	HIGH_USAGE_FLAG='Y'
email	EMAIL_TRIGGER='1'